Week 6 Worksheet

This is a worksheet for the week of Week 6. You do not need to turn this in.

Confidence Intervals for One Mean

When do we want to use confidence intervals?

- Without prior estimate for population parameter, use *confidence interval to estimate reasonable values for parameter*

How do we calculate confidence intervals?

Given a sample of *n* independent observations from an approximately normal distribution, the confidence interval for population mean μ is

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

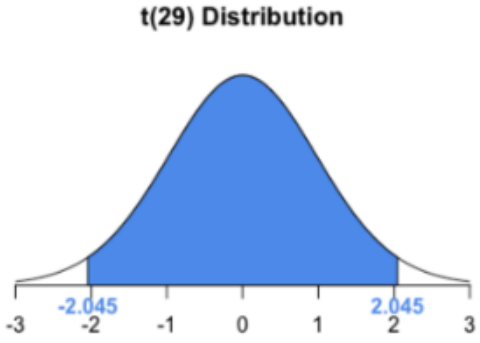
- $\underline{x}$ is the sample mean, *s* is the sample standard deviation
- t^* is the critical value for a particular confidence level and degrees of freedom
- *df = n − 1.*

How do we interpret the confidence interval?

The confidence level tells us how confident we can be that the interval we construct contains the true population mean. We say, we estimate with 95% confidence that the population mean of xyz is between (confidence interval).

How do we find the critical value t^*?

Like with our confidence intervals for *p*, typical choices for confidence levels are 90%, 95%, and 99%. To find the t^* associated with 95% confidence intervals with sample size of 30, we need to look at the t distribution with *df = n − 1 = 30 − 1 = 29*. Since we want an area of 0.95 between $-t^*$ and t^*, that leaves 1 – 0.95 = 0.05 to be split between the two tails.



Part I

Confidence Interval.

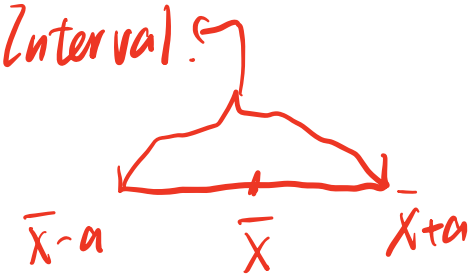
Population X. mean μ? sd: σ.

estimate?

draw sample of size n. $X_1, \cdots, X_n$. → calculate

Point Estimate:
- sample mean. $\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$
  $E(\bar{X}) = \mu$.
- sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$
  $E(S^2) = \sigma^2$.

Confidence Interval:



with high probability, population mean μ falls into the CI.

Part II. Approximate Normal Distribution.

① Population X is Normal mean μ sd: σ, sample size n.

Distribution of $\bar{X}$ is
- → normal. → mean $E(\bar{X}) = \mu$.
- → sd. $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.

$\therefore N(\mu, \frac{\sigma}{\sqrt{n}})$.

② Population X is not necessarily normal w. σ.

We can also say distribution of $\bar{X}$ is approximately normal if $n \geq 30$

⇓

Confidence Interval

$\bar{X} \pm t^* \times \frac{s}{\sqrt{n}}$ or $\bar{X} \pm z^* \times \frac{\sigma}{\sqrt{n}}$

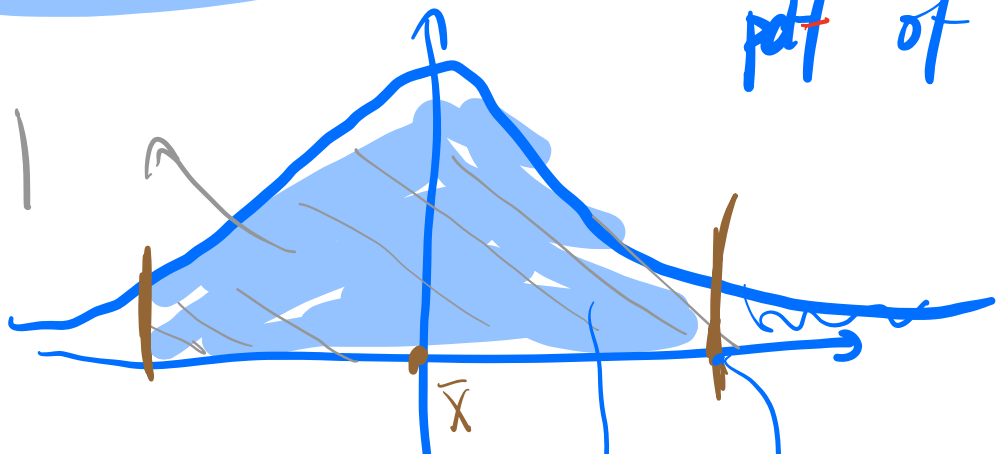
$\bar{x} -$ $\bar{x}$ $\bar{x} + t^* \cdot \frac{s}{\sqrt{n}}$

$z^* \cdot \frac{\sigma}{\sqrt{n}}$

t: when population sd: σ is unknown.

z: when population σ is known

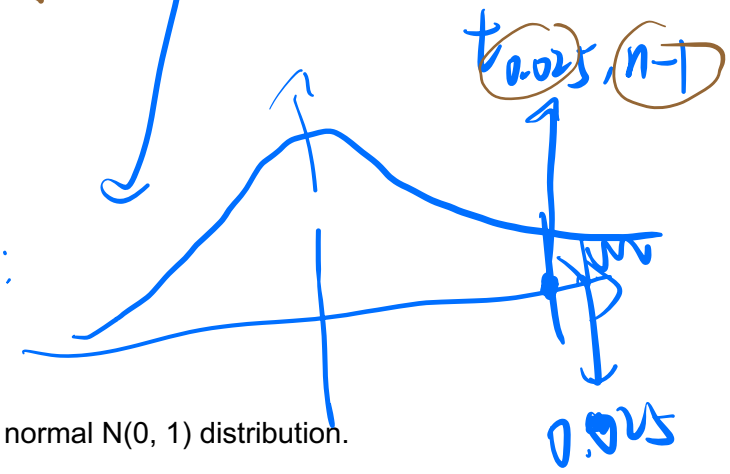
pdf of t dist.



95% CI.

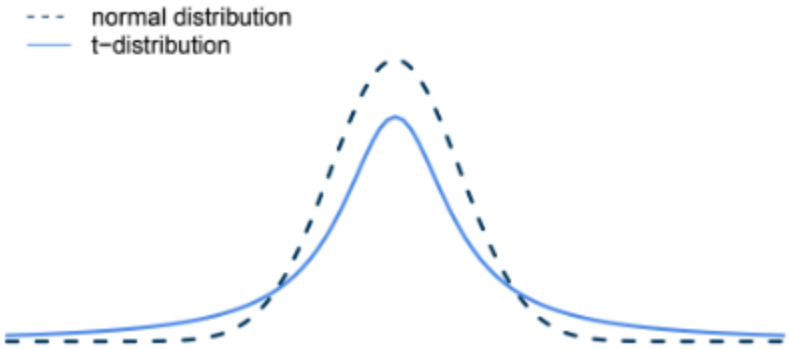
area : 0.95

t table :



Why are we using *t*? What happened to z*?*

The t distribution has thicker tails than the standard normal N(0, 1) distribution.



As a result, the critical value using the t distribution ($t^* = 2.045$) is larger than the critical value using the N(0, 1) distribution ($z^* = 1.960$). Since $t^* > z^*$ at a particular confidence level, confidence intervals using the t distribution are wider than confidence intervals using the N(0, 1) distribution.

This is the “penalty” we pay for using the sample standard deviation *s* to estimate the population standard deviation σ.

Question : Construct 95% CI.
population mean μ. sd: σ: Unknown.
sample of size n.

CI for mean μ:

Step 1: sample mean $\bar{X}$. sample var: S^2

Step 2: $\frac{1-0.95}{2} = 0.025$.
$n-1$ → paras for t dist.

Step 3: check t-table or $qt(u, v)$ function in r.
to find $t_{0.025, n-1}$

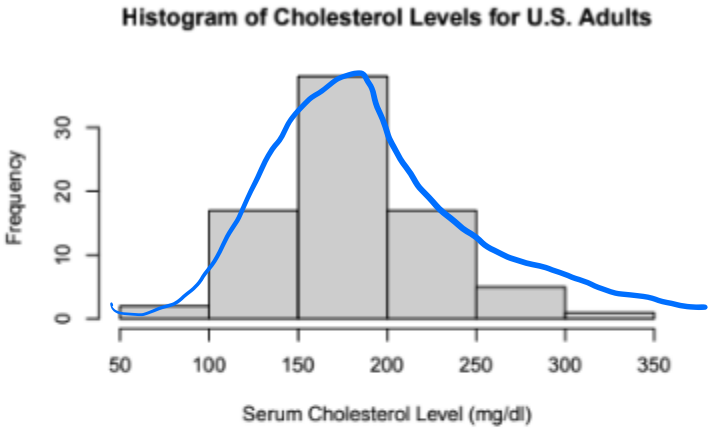
Step 4: 95% CI: $\bar{X} \pm t_{0.025, n-1} \times \frac{S}{\sqrt{n}}$

unknown

Example 1

The Centers for Disease Control and Prevention (CDC) reported that the **mean** serum total cholesterol level for adults aged 20 and older in the United States was **191 mg/dL** in 2015-2018.6 Recently, a researcher recruited a **random sample of 80** U.S. adults aged 20 and older to **estimate** the mean serum cholesterol level.

R code




```
summary(chol_levels)

##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   72.13  153.85  174.54  180.03  202.30  311.92
sd(chol_levels)

## [1] 45.19351
```

Step 1: Check conditions

The researcher plans to calculate a 95% confidence interval. Let μ represent the mean serum cholesterol level of U.S. adults aged 20 and over.

1. Independence: The independence condition is satisfied because the researcher took a random sample. Check the remaining condition for computing a confidence interval.
2. Normality: The histogram is unimodal and looks to be fairly symmetric, suggesting that the serum cholesterol levels came from a population with a normal distribution.

Step 2: Calculate the Confidence Interval

Use R to calculate the critical value:

```
qt(0.025, df = 79, lower.tail = FALSE)
```

```
## [1] 1.99045
```

$$df = n - 1 = 80 - 1 = 79$$

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}} = 180.03 \pm 1.990 \frac{45.19351}{\sqrt{80}} = 180.03 \pm 10.055 = (169.975, 190.085)$$

Step 3: Interpretation

What is wrong with these interpretations?

Interpretation 1: We estimate with 95% confidence that the sample mean serum cholesterol level of U.S. adults aged 20 and over is between 169.975 and 190.085 mg/dl.

Interpretation 2: 95% of the serum cholesterol levels of U.S. adults aged 20 and over is between 169.975 and 190.085 mg/dl.

Interpretation 3: There is a 95% chance that the population mean serum cholesterol level of U.S. adults aged 20 and over is between 169.975 and 190.085 mg/dl.

Correct Interpretation:

We estimate with 95% confidence that the population mean serum cholesterol control level of US adults aged 20 and over is between 169.975 and 190.085.

How does Confidence Interval and Hypothesis Test relate?

There is a relationship between confidence intervals and hypothesis tests. Consider a 95% confidence interval and a test of $H_0 : \mu = \mu_0$ and $H_A: \mu \neq \mu_0$ at the $\alpha = 0.05$ significance level.

- Any μ_0 value within the 95% confidence interval is a value that would result in a p-value > 0.05, meaning we would fail to reject the null hypothesis
- Any μ_0 value outside of the 95% confidence interval is a value that would result in a p-value < 0.05, meaning we would reject the null hypothesis

Example 2

Would you reject or fail to reject H_0: $\mu = 170$ and H_A: $\mu \neq 170$ at the $\alpha = 0.05$ significance level?

Would you reject or fail to reject H_0: $\mu = 195$ and H_A: $\mu \neq 195$ at the $\alpha = 0.05$ significance level?

Since 170 is in the confidence interval, it's a reasonable value for the population mean. We would fail to reject the null hypothesis.

Since 195 is not in the CI, we don't think it's a reasonable value for population mean. We would not fail to reject the null hypothesis.